

Sample questions on Bagging and Boosting

Roll: _____

Self practice

Name: _____

Mode: Credit/Audit _____

Write all your answers in the space provided. Do not spend time/space giving irrelevant details or details not asked for. You are expected to solve most of the questions on your own. If you discuss any solution with someone mention alongside that question the person with whom you discussed it.

- Suppose we use Bagging to generate a set of B independent regression functions $f_1(\mathbf{x}), \dots, f_B(\mathbf{x})$. The prediction of the bagged regression function $F(\mathbf{x}) = \frac{1}{B} \sum_j f_j(\mathbf{x})$.

- If σ^2 denotes the expected square error $E((y - f_j(\mathbf{x}))^2)$ of each function $f_j(\mathbf{x})$, what is the square error of $F(\mathbf{x})$ assuming that the errors of different classifiers are independent of each other. ..3 $E((y - F(\mathbf{x}))^2) = ((y - \frac{1}{B} \sum_j f_j(\mathbf{x}))^2) = E(\frac{1}{B^2}(\sum_j (y - f_j(\mathbf{x}))^2 - 2 \sum_{j,k} (y - f_j(\mathbf{x}))(y - f_k(\mathbf{x}))))$

The first term in the expectation is $\frac{\sigma^2}{B}$. The second term equals zero because the different bags are independent of each other and have zero correlation.

- Suppose in the training set there is a severe outlier $(\mathbf{x}^r, y_r)$ that causes all functions that includes it to make correlated mistakes. What is the probability δ that a given component f_j will encounter this outlier in its training set? (Recall that the training set of f_j will be obtained by sampling with replacement N instances randomly from the original training set D)? ..3 The probability δ that the instance will be included in the sample of bag b is $1 - (1 - \frac{1}{N})^N$
- Let S be the set of component functions that include the outlier in their training sample. All functions in S have a higher square error of β^2 . Also, there is a correlation c in the errors made by any two component functions from the set S . For other pairs of functions there is no correlation. Now, what is the expected error of $F(\mathbf{x})$. ..3

- Mrinal is training a regression tree with gradient boosting. The first weak learner he obtained is given by

$$F_1 = 1.6 * \mathbb{I}(x < 2) + 5.3 * \mathbb{I}(x \geq 2)$$

. He is using a clipped L-1 loss given below.

$$L(y_i, F(x^i)) = \begin{cases} 0.6 & : |y_i - F(x^i)| > 0.6 \\ |y_i - F(x^i)| & : \text{otherwise} \end{cases}$$

Given the dataset in the table, help him fill the values of residuals and gradient of the loss in the missing entries.

x	y	Residual	Gradient
1	1.2	-0.4	-1.0
1.7	2.1	0.5	1.0
3.8	4	-	-
4.3	5.7	-	1.0
4.6	5.4	0.1	-
5.2	6.3	1.0	0.0

3. Consider a two class training dataset D with $d = 1$ attributes where a fraction ϵ of the points have x value -2 and are labeled positive (+1), a fraction $(1 - \epsilon)/2$ are uniformly distributed between 0 and α where $\alpha > 0$ and are labeled negative (-1), and the remaining $(1 - \epsilon)/2$ are uniformly distributed between α and 2α and are labeled positive. Suppose we run boosting on this dataset where each stage is a decision tree classifier restricted to have a single node. Assume the decision tree output 1 or -1 . Draw the decision tree of the first two stages of running Gradient boosting with binary cross entropy loss. The split condition is to specified in terms of α and ϵ . ..4

Call instances at -2 , group A instances, negative instances group B, and rest group C.

- First tree: weight of all instances 1 , split at $x_1 < \alpha$, group A instances misclassified, the rest are correctly classifier. The gradient of the loss will be 1 for group A, and 0 for the rest.
- Second tree: weight of group A $= 1$.
Split at $x_1 < 0$.